

AI 2002

Artificial Intelligence

Lecture 16

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Learning

- An agent is **learning** if it improves its performance on future tasks after making observations about the world.
- **Learning** is the ability of an agent to improve its behavior based on experience.

Learning

- This could mean the following:
 - The **range of behaviors** is expanded; the intelligent agent can do more.
 - The **accuracy level to perform tasks** is improved; the intelligent agent can do things in a better way.
 - The efficiency in terms of **speed** is improved; the intelligent agent can do things faster.

Example:

When to drive the car?

It depends on,

- Temperature
- Expected precipitation
- Day of the week
- Whether need to shop on the way back home
- What are you wearing

Memory

Temp	Precip	Day	Shop	Cloths	
80	None	Sat	No	Casual	Walk
19	Snow	Mon	Yes	Casual	Drive
65	none	Tue	No	Casual	Walk
19	Snow	Mon	Yes	Casual	??

Averaging

Temp	Precip	Day	Shop	Cloths	
80	None	Sat	No	Casual	Walk
80	None	Sat	No	Casual	Walk
80	None	Sat	No	Casual	Drive
80	None	Sat	No	Casual	Drive
80	None	Sat	No	Casual	Walk
80	None	Sat	No	Casual	Walk
80	None	Sat	No	Casual	Walk
80	None	Sat	No	Casual	?

Generalization

Temp	Precip	Day	Shop	Cloths	
71	None	Fri	Yes	Casual	Drive
36	None	Sun	Yes	Casual	Walk
62	Rain	Weds	No	Casual	Walk
93	None	Mon	No	Casual	Drive
55	None	Sat	No	Formal	Drive
80	None	Sat	No	Casual	Walk
19	Snow	Mon	Yes	Casual	Drive
65	None	Tues	no	Casual	Walk

Generalization

- He's going to **walk** because it's raining today and the only other time it rained, he walked.
 - He's going to **drive** because he has always **driven on Mondays**.
 - He's going to **walk** because he only drives if he is wearing formal clothes, or if the temperature is above 90 or below 20.
- The question of which one to choose is hard.

Learning Types

- Learning may be:
 - Supervised Learning
 - Unsupervised Learning
 - Semi-supervised Learning
 - Reinforcement Learning

Supervised Learning

- In **supervised learning**, the agent observes some example input–output pairs and learns a **function that maps from input to output**. Supervised learning involves:
 - input features
 - target features
 - training examples

Supervised Learning

where the input features as well as the target features are specified.

- We have to predict **the target features** of a new example for which the input features are given.
- This is called,
 - **classification** when the target variables are discrete and
 - **regression** when the target features are continuous.

Supervised Learning

- Given a data set (training data)

$$D = (x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)$$

Goal: Find a hypothesis h in hypothesis class H that performs a good job of mapping x to y .

When y_i is a Boolean, or a member of a discrete set, the problem is a **classification** problem. When y_i is real-valued, we call this a **regression** problem.

Error Measure for Classification

- Y is discrete, a (small) finite, unordered set of classes

$$\text{error}(h(x), f(x)) = 0 \text{ if } h(x) = f(x) \text{ else } 1$$

0-1 Loss Error

Error Measure in Regression

Y is continuous, a numeric set (typically real numbers)

$$\text{error } (h(x), f(x)) = (h(x) - f(x))^2$$

Squared Error

Unsupervised Learning

- In **unsupervised learning**, the agent learns patterns in the input even though **no explicit feedback** is supplied.
- **Unsupervised learning** occurs when **no classifications are given and the learner must discover categories and regularities in the data.**

Unsupervised Learning Example

- The most general example of unsupervised learning task is **clustering**:
 - potentially useful clusters developed from the input examples.
- For example, **a taxi agent** might gradually develop a concept of “good traffic days” and “bad traffic days”.

Semi-Supervised Learning

- In **semi-supervised learning** we are given a few labeled examples and must make what we can of a **large collection of unlabeled examples**.
- Some data is labeled but most of it is unlabeled, and a mixture of supervised and unsupervised techniques can be used.
- Many real-world machine learning problems fall into this type of learning.

Reinforcement Learning

- A supervised learning agent needs to be told the correct move for each position it encounters, **but such feedback is seldom available.**
- In the absence of feedback, **an agent can learn a transition model for *its own moves*** and can perhaps learn to predict the opponent's moves,
- *Without **some feedback about what is good and what is bad**, the agent will have no grounds for deciding which move to make.*

Reinforcement Learning

- In **reinforcement learning** the agent learns from a series of reinforcements—**rewards or punishments**.
- A win at the end of a chess game **tells the agent it did something right**.
 - It is up to the agent to decide which of the **actions prior to the reinforcement were most responsible for it**.
- The rewards **may come more frequently**, it depends upon the environment.

Reinforcement Learning

- Each percept(e) is enough to determine the State(the state is accessible)
- The agent can decompose the reward component from a percept.
- The **agent task**: *to find an optimal policy, mapping states to actions, that maximize a long-run measure of the reinforcement*
 - Think of reinforcement as a reward